



University
of Windsor

Comp-3150 (60-315) Winter 2020

Midterm 1 (for practice only 2026)

Examiner: Dr. C. I. Ezeife Given: Thursday, Oct. 8, 2020

Student Name: _____

Student Number: _____

INSTRUCTIONS (Please Read Carefully)

Examination Period is 1 hours 20 minutes

Answer all questions. Write your answers in the spaces provided in the question paper. This is closed book and closed notes test.

Total Marks =50. Total number of sections = 2

Please read questions carefully! Misinterpreting a question intentionally or unintentionally results in getting a “ZERO” for that question. Good Luck!!!

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CONFIDENTIALITY AGREEMENT & STATEMENT OF HONESTY

I confirm that I will keep the content of this examination confidential.

I confirm that I have not received any unauthorized assistance in preparing for or writing this examination. I confirm knowing that a mark of 0 may be assigned for copied work.

Student Signature

Student Name (please print)

Student I.D. Number

Date

For marking purposes only (This part not to be filled by students)

Question	Mark
Section A (15 marks for 10 multiple choice questions)	
	/15
Section B	
Que 1 (15 marks)	
	/15
Que 2 (10 marks)	
	/10
Que 3 (10 marks)	
	/10
Total	/50

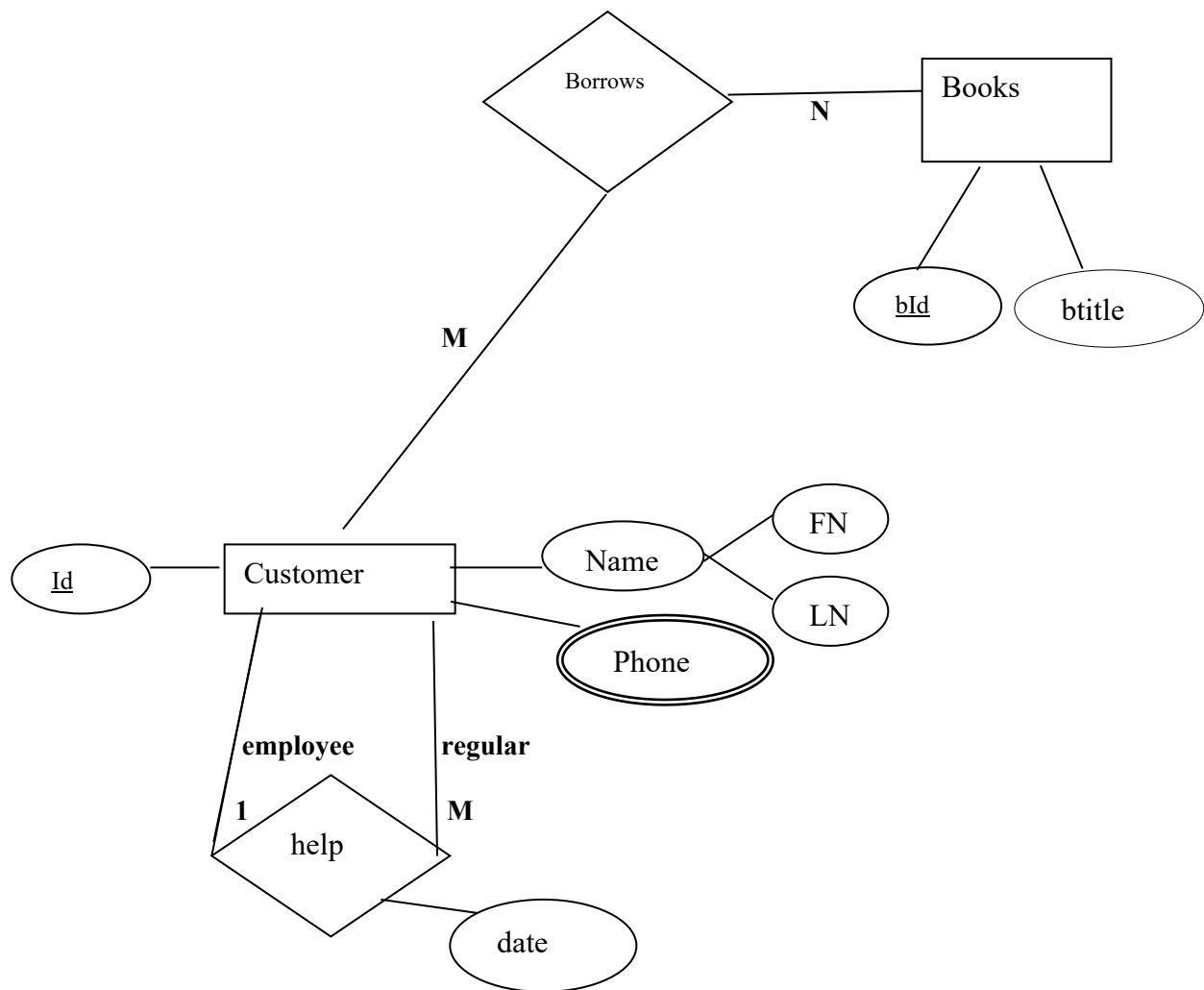
Section A

15 marks for 10 Multiple Choice Questions. Each question in this section is worth 1.5 marks.

For Questions 1 – 10, circle the correct answer .

1. Recall that a database has many types of users, each of whom may require a different view of the database. The database user responsible for defining data to be stored in the database given the requirements description is called the:
 - a. The database administrator (DBA)
 - b. Database end user
 - c. DBMS system builder
 - d. Database designer
 - e. none of the above
2. Which of the following can be an example of a composite attribute in ER modeling of a database?
 - a. address
 - b. last name
 - c. first name
 - d. country
 - e. none of the above
3. Some of the main advantages of using a database management system as opposed to the traditional file system of running programs with data is:
 - a. Using programming languages such as JAVA to create database applications
 - b. Separation of data from program using the three data level architecture
 - c. Replication of data and programs
 - d. Use of the internet
 - e. none of the above
4. In a database, integrity constraints must not be violated after any database update, given a database table with schema Customer (Id, Fname, Lname, phone), which has a state having the two tuples <1, 'Mark', 'Adam', '416-555-2233'> and <2, 'Pat', 'Moore', '519-333-6161'>, what integrity constraint is violated by inserting the following new tuple into Customer <NULL, 'Ken', 'Hardy', '226-222-1234'>.
 - a. Semantic Integrity constraint
 - b. Foreign key referential integrity constraint
 - c. Primary Key integrity constraint
 - d. All of the above
 - e. none of the above

Figure A.1: ER diagram Customer-borrows-books Database

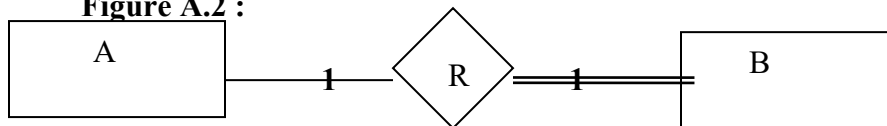


Use the ER diagram of Figure A.1 to answer the next few questions

5. Which statement best describes the cardinality constraint of the relationship **Borrows** shown in Figure A.1 above.
 - a. Each customer can borrow only one book, and each book can be borrowed by many customers.
 - b. Each customer can borrow many books, and each book can be borrowed by many customers.
 - c. Each customer can borrow many books, and each book can be borrowed by only one customer.
 - d. All of the above.
 - e. None of the above.
6. Which statement best describes the participation constraint of the relationship **'help'** shown in Figure A.1 above.

- a. Every employee customer must help some regular customers through ‘help’ recursive relationship.
 - b. Every regular customer must be helped by some employee customer through ‘help’ recursive relationship.
 - c. Not every employee customer ought to have helped a regular customer through ‘help’ recursive relationship.
 - d. All of the above.
 - e. None of the above.
7. Which statement best describes the entities and relationships in Figure A.1 above?
- a. Customer is a weak entity.
 - b. Every employee customer ought to have helped some customers to be in the database.
 - c. Phone of a student is a composite attribute.
 - d. All of the above.
 - e. None of the above.

Figure A.2 :



8. Which statement best describes the relationship shown in Figure A.2 above.
- a. Each A record can relate to only one record of B through R.
 - b. Each A record can relate to multiple records of B through R.
 - c. Each B record can relate to multiple records of A through R.
 - d. There can be multiple records of A and B related through R.
 - e. None of the above
9. In a database, the 3-level data architecture allows us to define the simple database “to track books borrowed by customers” using the dbms data model at the conceptual level. An external level query view of this database is:
- a. Print all customers who borrowed a book
 - b. Print all customer names and their phones
 - c. Print all book titles borrowed by a specific customer
 - d. All of the above
 - e. None of the above
10. Using “the customer borrows books” database described in Fig A.1, when the structure of the book file/table in the database is changed at the conceptual level to include more attributes, the external view queries do not need to change and this is because:
- a. the dbms provides logical data independence
 - b. the dbms provides physical data independence
 - c. the dbms provides concurrent access to data
 - d. the dbms provides a query language for processing data
 - e. None of the above

Section B (35 marks):
This section has 3 questions :

1. **(15 marks)** Given the ER diagram of the COMPANY database of Figure 3.2, answer the following questions with this diagram.

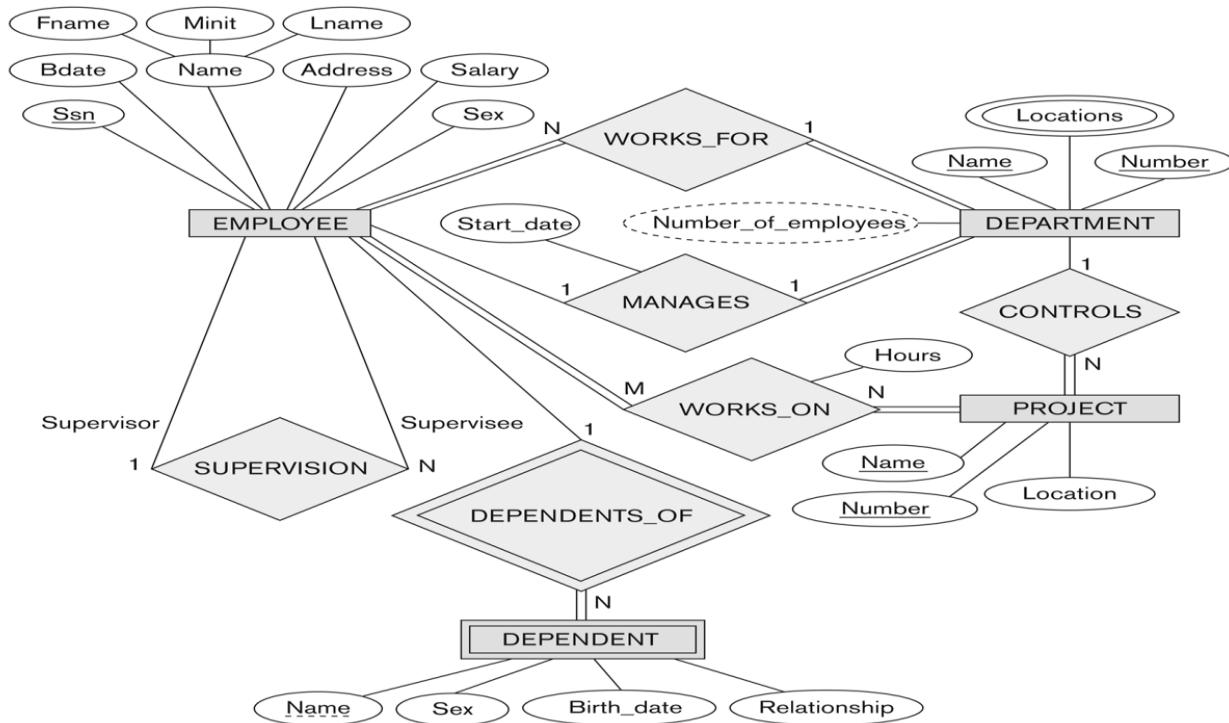


Figure 3.2
 An ER schema diagram for the COMPANY database. The diagrammatic notation is introduced gradually throughout this chapter.

Write the relational database schema (in text form) for COMPANY including all the entities with attributes, relationships with attributes and list at least two of the six cardinality and participation constraints that describe each of the six relationships in this database (Full description of one relationship constraints stands for one).

COMPANY Database Component	COMPANY SCHEMA for the Database Component
Entities and attributes (5 marks)	

Relationships and attributes (5 marks)	
Cardinality and Participation Constraints (5 marks)	

2. **(10 marks)** Given the following relational database schema for a phone subscription database application, answer the questions that follow.

CUSTOMER (SSN, Cname, City)
PHONE(Phonetype, PhoneMaker)
PHONECOMPANY (CompId, CompName, Numproducts)
SUBSCRIBETO (SSN, CompId, Phonetype, Month, Billamt)

Here, Numproducts refers to the number of phone products carried by the phone company.
Billamt is the dollar amount of bill for the month on the phone to the customer with SSN.

- a. Specify all the foreign keys for this schema. State each foreign key using the following format "foreign key is the attribute ----- of relation ----- that references relation -----".
(5 marks)

- b. State two queries involving more than one table that can be posed on this database indicating the tables to visit to answer the queries each time. **(5 marks)**

Question	Foreign Key
2a (5 marks)	

Question	Query	Tables needed to answer query
2b (5 marks)		

3. (10 marks)

- a. Design a simple STUDENT records database with not more than 3 tables **(5 marks)**
- b. Create an instance of your database in (a). **(5 marks)**

Question	Answers
a. (5 marks)	

b. (5 marks)	
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